

OCNA SUGATAG, Romania

WMO: 150150

Lat: 47.7769N

Lon: 23.9406E

Elev: 504

StdP: 95.41

Time Zone: 2.00 (EUE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-10.6	-18.8	0.8	-10.9	-15.6	1.0	-8.9	6.3	2.7	5.2	2.1	1.1	190	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	30.0	19.9	28.2	19.4	26.6	18.5	21.2	27.5	20.2	26.5	19.3	25.2	1.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.5	24.6	18.0	13.7	23.3	17.1	13.0	22.2	63.9	27.5	60.3	26.6	57.2	25.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.3	3.6	-16.0	32.3	2.5	1.7	-17.8	33.5	-19.3	34.5	-20.7	35.4	-22.6	36.6		
(5)			-16.5	22.6	2.5	0.7	-18.3	23.1	-19.7	23.5	-21.1	23.9	-22.9	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-1.9	0.1	4.2	10.3	14.3	17.7	19.1	19.4	14.6	9.8	5.3	-0.2		
	DBStd	8.44	4.70	4.77	4.61	3.72	3.47	3.22	2.94	3.08	3.75	3.96	4.64	4.51		
	HDD10.0	1404	368	277	186	40	6	0	0	0	6	54	152	316		
	HDD18.3	3396	626	511	439	240	129	49	27	25	119	265	391	574		
	CDD10.0	1202	0	0	4	51	141	230	281	290	146	48	11	0		
	CDD18.3	151	0	0	0	1	6	29	50	57	8	1	0	0		
	CDH23.3	1434	0	0	0	11	72	258	441	551	99	2	0	0		
	CDH26.7	356	0	0	0	1	4	44	118	169	19	0	0	0		
(14) Wind	WSAvg	1.4	1.1	1.2	1.7	1.7	1.6	1.5	1.5	1.4	1.4	1.3	1.3	1.1		
(15) Precipitation	PrecAvg	858	50	49	57	59	90	99	103	79	80	71	61	60		
	PrecMax	1115	123	145	153	91	199	175	228	235	155	168	134	106		
	PrecMin	652	22	19	9	21	25	35	21	19	19	8	5	9		
	PrecStd	139	22	26	38	21	35	40	58	43	35	42	30	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.9	18.0	25.0	27.2	29.8	31.7	32.7	29.0	23.1	18.6	11.8		
		MCWB	6.7	7.9	10.0	14.2	16.5	20.0	20.3	19.7	18.3	14.7	12.5	7.9		
	2%	DB	7.4	9.9	15.3	21.5	25.4	27.9	29.7	30.5	25.9	20.7	15.7	9.9		
		MCWB	5.5	5.8	8.7	12.3	16.0	19.3	20.2	19.8	16.9	13.9	10.8	7.1		
	5%	DB	6.0	7.9	13.1	19.6	23.5	26.3	27.7	28.5	23.7	18.4	13.6	7.9		
		MCWB	4.5	5.0	7.4	11.2	15.7	18.6	19.5	19.5	16.2	12.9	9.8	5.7		
	10%	DB	4.2	6.2	11.0	17.5	21.4	24.6	25.7	26.5	21.5	16.2	11.8	6.0		
		MCWB	2.9	4.0	6.5	10.3	14.7	17.7	18.7	18.9	15.3	11.8	8.8	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.1	10.8	14.8	18.3	21.9	22.3	22.5	19.1	15.7	13.1	9.0		
		MCDB	8.5	12.0	16.2	22.8	24.7	28.2	28.7	29.3	26.8	21.0	16.8	10.9		
	2%	WB	5.9	6.6	9.4	13.3	17.1	20.4	21.1	21.2	17.8	14.5	11.6	7.3		
		MCDB	7.1	9.2	14.2	20.6	23.2	26.3	27.5	27.6	24.0	19.3	14.8	9.6		
	5%	WB	4.7	5.5	8.1	12.0	16.1	19.3	20.1	20.1	16.9	13.5	10.3	5.9		
		MCDB	5.8	7.7	12.1	18.0	22.1	24.8	26.6	26.5	22.5	17.5	12.9	7.9		
	10%	WB	3.1	4.3	6.8	10.8	15.2	18.3	19.2	19.2	15.8	12.3	9.1	4.4		
		MCDB	4.1	6.1	10.3	16.1	20.6	23.2	24.9	25.3	20.7	15.7	11.7	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.7	8.5	10.3	10.7	10.8	11.0	11.7	10.5	9.7	7.2	5.2		
		MCDBR	6.3	8.6	12.0	13.7	14.0	13.4	14.0	14.9	14.1	12.6	10.0	7.4		
		MCWBR	4.7	5.6	6.6	6.5	6.7	6.3	6.3	6.1	6.5	6.7	5.8	5.3		
	5% WB	MCDBR	5.5	7.9	10.6	12.1	12.7	12.4	12.9	13.3	12.6	11.2	8.7	7.0		
		MCWBR	4.2	5.3	6.0	6.1	6.3	6.2	6.2	6.1	6.2	6.1	5.2	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.340	0.387	0.441	0.424	0.419	0.442	0.443	0.402	0.370	0.342	0.313		
	taud		2.254	2.191	2.149	2.081	2.200	2.247	2.194	2.188	2.261	2.322	2.333	2.313		
	Ebn at Noon		745	801	818	808	841	846	819	799	796	762	705	697		
	Edh at Noon		78	104	127	150	139	133	139	134	113	91	73	66		
(44) All-Sky Solar Radiation	RadAvg		1.07	1.89	3.04	4.29	5.26	5.66	5.58	5.13	3.61	2.35	1.28	0.81		
	RadStd		0.16	0.26	0.43	0.45	0.55	0.41	0.51	0.47	0.47	0.32	0.18	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.70	N/A	N/A	+1.04	N/A	N/A	-150	-216	+105	+39

Nomenclature: See separate page